

DATE:

EXPT NO:03

EDA-Data Inspection and Analysis

AIM:

To inspect and analyze the Earthquake dataset using Exploratory Data Analysis (EDA) to understand the data structure, identify patterns, detect missing values, and discover relationships between different earthquake attributes.

PROCEDURE:

1. Import Pandas, NumPy, Matplotlib, and Seaborn libraries.
2. Load the Earthquake dataset using Pandas.
3. Display the first and last records using head() and tail().
4. Check the number of rows and columns using shape.
5. Display column names and data types using info().
6. Check missing values using isnull().sum().
7. Generate statistical summaries using describe().
8. Check and identify duplicate records.
9. Analyze important earthquake-related attributes.
10. Create suitable graphs such as histograms, bar charts, and scatter plots.
11. Generate a correlation heatmap to identify relationships between numerical variables.
12. Interpret the findings and identify important patterns in the Earthquake dataset.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

file_path = r"C:\Users\babyn\Downloads\archive\earthquake_data.xlsx"
df = pd.read_excel(file_path)
print("Dataset Preview:")
print(df.head())

# 1. Identifying Missing Values
print("\nMissing Values in Each Column:")
print(df.isnull())

# 2. Count Missing Values
print("\nTotal Missing Values:")
print(df.isnull().sum())

# Import required libraries
import pandas as pd
```

```

import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler

# Load Dataset
file_path = r"C:\Users\babyn\Downloads\archive\earthquake_data.xlsx"
df = pd.read_excel(file_path)
print("Original Dataset:")
print(df.head())
print("\nDataset Information:")
print(df.info())

# 1. Identify Missing Values
print("\nMissing Values in Dataset:")
print(df.isnull())

# 2. Count Missing Values
print("\nTotal Missing Values in Each Column:")
print(df.isnull().sum())

# 3. Replace Missing Values
# Numerical columns - replace missing values with mean
numeric_columns = df.select_dtypes(include=np.number).columns
for col in numeric_columns:
    df[col] = df[col].fillna(df[col].mean())

# Categorical columns - replace missing values with mode
categorical_columns = df.select_dtypes(include="object").columns
for col in categorical_columns:
    df[col] = df[col].fillna(df[col].mode()[0])

print("\nMissing Values After Replacement:")
print(df.isnull().sum())

# 4. Detect Duplicate Records
print("\nNumber of Duplicate Rows:")
print(df.duplicated().sum())

# Remove duplicate records
df = df.drop_duplicates()
print("\nDuplicate Rows After Removing:")
print(df.duplicated().sum())

# 5. Detect Outliers using IQR Method
print("\nOutlier Detection:")
outlier_result = {}
for col in numeric_columns:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - (1.5 * IQR)

```



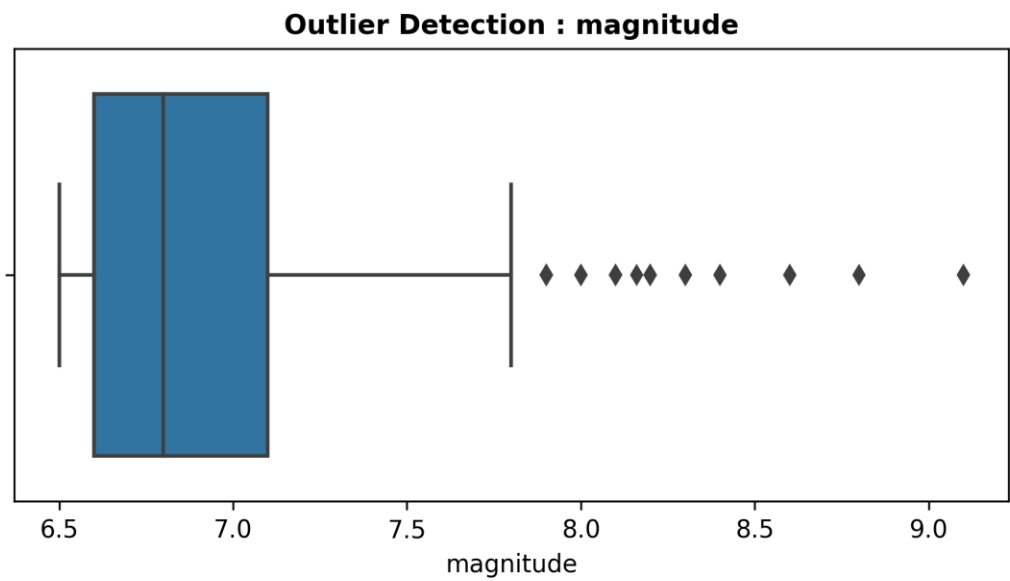
```
2 green          1 670 us 131 4.998 27.0 mww 624.464 -25.5948 178.278
NaN            NaN          NaN
```

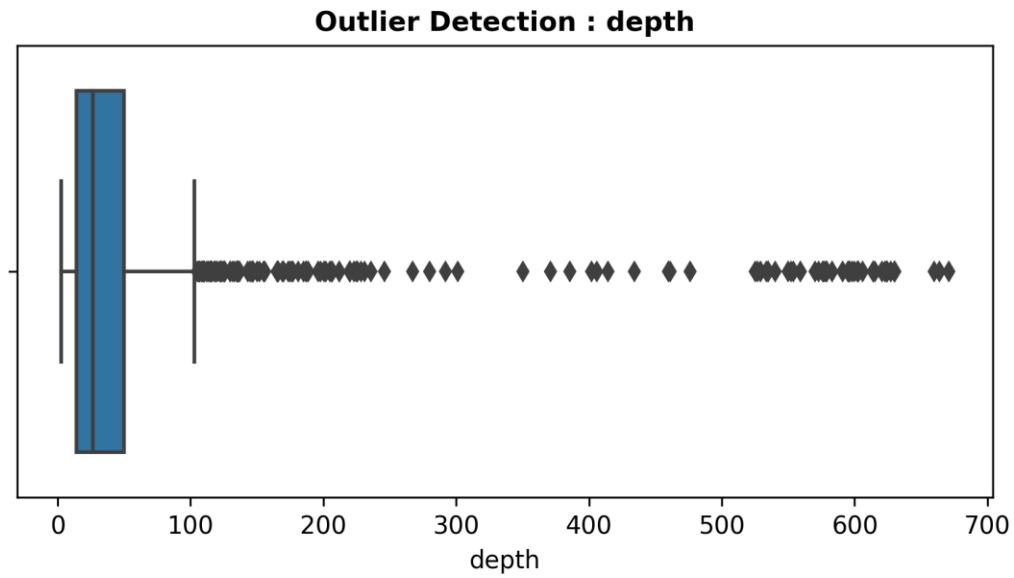
Missing Values in Dataset:
Total Missing Values in Each Column:

title	0
magnitude	0
date_time	0
cdi	0
mmi	0
alert	367
tsunami	0
sig	0
net	0
nst	0
dmin	0
gap	0
magType	0
depth	0
latitude	0
longitude	0
location	5
continent	576
country	298

dtype: int64

Outlier Detection:
{ 'magnitude': 38, 'cdi': 0, 'mmi': 0, 'tsunami': 0, 'sig': 42, 'nst': 34, 'dmin': 48, 'gap': 25, 'depth': 86, 'latitude': 0, 'longitude': 0 }





RESULT:

The Earthquake dataset was successfully inspected and analyzed using EDA. The dataset structure, missing values, duplicates, statistical details, and important relationships were identified through data analysis and visualization.